

Raza Shaud

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SUMMARY

Electrical Engineering graduate (B.Tech, June 2026) and GATE 2026 qualifier (Score: 515, AIR: 2139) with strong expertise in machine learning, computer vision, and embedded systems. Delivered 4 technical projects achieving 96.5% average accuracy and automating 99% of manual processes. Completed 3 industrial internships across power generation and steel manufacturing sectors. Published 3 research papers including an IEEE Best Paper Award and an international conference presentation on intelligent power systems.

TECHNICAL SKILLS

- **Programming Languages:** Python | C | C++
- **AI/ML & Data Science:** Deep Learning | CNN | Scikit-learn | NumPy | Pandas | Random Forest
- **Computer Vision & IoT:** OpenCV | YOLOv5 | Arduino IDE
- **Engineering Software:** ETAP | ANSYS | MATLAB | Microsoft Office

PROFESSIONAL EXPERIENCE

Tech Solutions Inc. | ML Intern

May 2025 – July 2025

- Automated data pipelines for 5,000+ records, cutting preparation time by 60% & Optimized 8 ML scripts, boosting prediction speed by 35%.
- Tested models with a 4-person team, achieving 92% accuracy across 3 use cases.

Steel Authority of India Limited (SAIL) | Internship Trainee

Nov 2024 – Dec 2024

- Observed industrial automation processes and electrical systems in a large-scale steel manufacturing environment.

Damodar Valley Corporation (DVC) | Internship Trainee

June 2024 – July 2024

- Observed and documented daily operational routines with senior engineers at a major thermal power plant and gained hands-on exposure.

EDUCATION

Bachelor of Technology in Electrical Engineering | Ghani Khan Choudhury Institute of Engineering and Technology

Nov 2022 – Jul 2026

KEY PROJECTS

Solar-Powered Robotic Cable Monitoring System | Arduino, Solar PV, CNN, OpenCV, Python

- Engineered autonomous mobile robot integrating 12V solar PV system with camera-based cable insulation monitoring across 500m test infrastructure, eliminating 100% manual inspection requirements.
- Architected CNN-based fault detection model processing 1300+ cable images achieving 96.5% accuracy, reducing false positives by 78% compared to traditional visual inspection methods.
- Optimized power consumption achieving 8-hour continuous operation on 20W solar panel, cutting operational costs by ₹15,000 annually per monitoring unit.

Human vs AI Voice Classification System | Python, Random Forest, MFCC

- Developed binary classification system analyzing 1500+ audio samples using MFCC feature extraction to distinguish authentic human voice from AI-generated speech with 89% accuracy.
- Implemented Random Forest model with 15-dimensional feature vector, outperforming baseline SVM by 12% through hyperparameter tuning and cross-validation across 5 folds.
- Preprocessed audio dataset reducing noise interference by 65% using spectral gating techniques, improving model generalization on real-world recordings.

Face Recognition Attendance System | Python, OpenCV, Pandas

- Built real-time facial recognition attendance system processing 30 faces per second, automating attendance logging for 100+ students and reducing manual record-keeping by 99%.
- Integrated face encoding database with Pandas-based CSV logging achieving 95% recognition accuracy under varying lighting conditions across 200+ daily scans.
- Decreased attendance processing time from 15 min to 45 sec per session, saving 28 instructor-hours per semester.

Saline Monitoring and Alert System | Arduino Uno, Ultrasonic Sensor, GSM Module

- Designed IoT-based medical monitoring system tracking saline levels with 2cm precision using ultrasonic sensors, triggering automated alerts to prevent 100% of empty-bottle incidents during 50-hour testing.
- Programmed dual-alert mechanism combining buzzer notifications and GSM-based SMS to 3 medical staff contacts, reducing response time by 75% in clinical simulation environments.
- Reduced patient monitoring workload by 40% for nursing staff through real-time automated surveillance.

PUBLICATIONS

- "Characteristics Investigation of Dual Rotor Single Stator 6-Phase Single I-Shaped Magnetic Pole PMSM for Smart Electric Vehicles" | IEEE GIEST 2024 International Conference | Best Paper Award
- "Noise Filtration and Voice Discrimination: Case Studies, Results, and Discussions" | Tech Research Chronicles (IAST) | Vol. 1, Issue 2, 2025 | ISSN: 3049-205X (Print)
- "Intelligent Remote Monitoring of Low Voltage Cable Insulation Fault" | PEIS-2026 | April 3–4, 2026 | GKCIET, Malda

AWARDS & CERTIFICATIONS

- GATE 2026 | Electrical Engineering (EE) | Score: 515 | Marks: 42.0 / 100 | AIR: 2139 (UR)
- 100 Days of Code: The Complete Python Pro Bootcamp, Udemy | 2024